Portfolio Risk Measure and Diversification

What is Risk?

- Risk of losing money
- Risk of underperforming benchmark/peer group
- Risk of being under-funded
- Value at Risk, Shortfall probability
- Uncertainty of returns

Distributions and Risk Measures

Distributions

- Distribution describes the probability of all possible outcomes
- Distributions answer all questions about returns and probability
- Distributions are very complex objects
- A characteristic of a distribution might serve as a measure for risk but it will always be a simplification

Risk measures

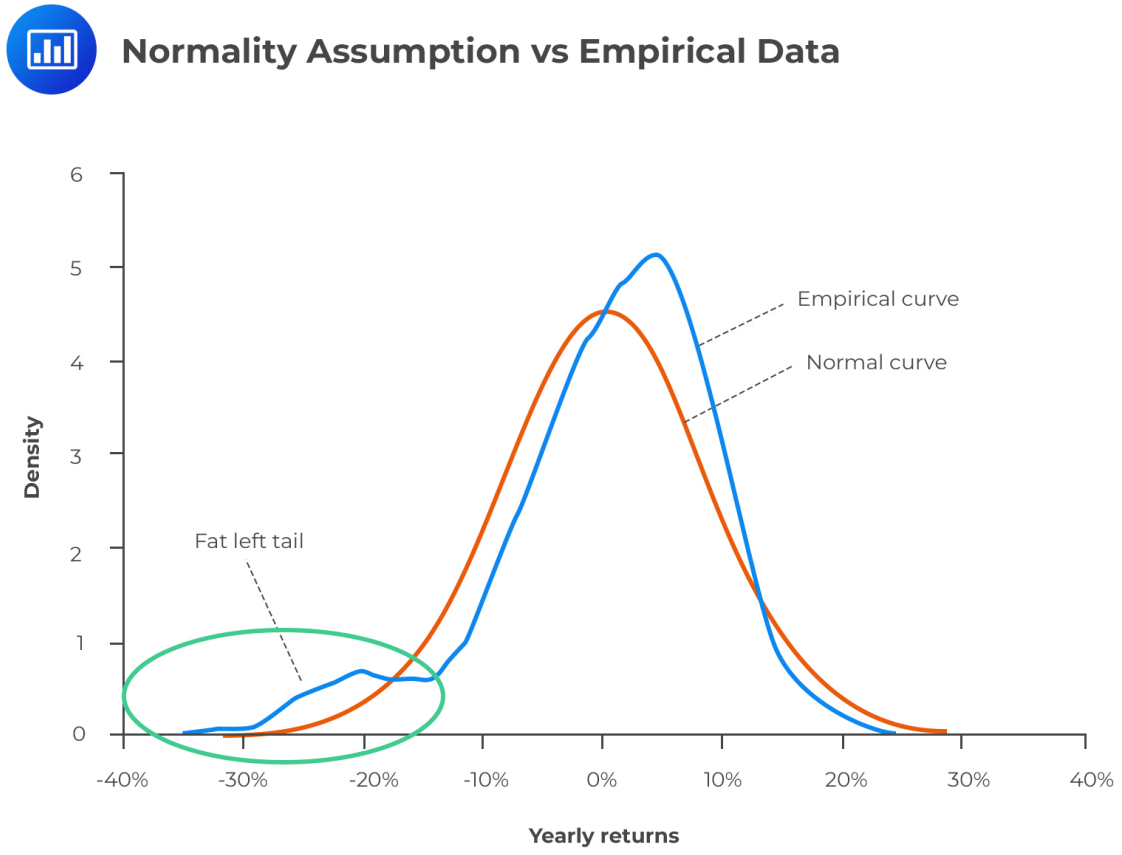
- Look for an easy and robust concept to measure uncertainty
- The statistical properties of a risk measure should be well known
- We want a risk measure that allows us to calculate the portfolio risk from the risk of the underlying assets

Volatility – A measure of dispersion

- Using historical return data, we can estimate the variance of the return series as $$\sigma^2 = \frac{1}{n}\sum_{t=1}^{n}(r_t - \mu)^2$$
- The standard deviation of the return series is the square root of the variance
- The volatility of an asset is defined as the standard deviation of returns
- Volatility quantifies the width of the distribution.
- Volatility does not capture the full complexity of the distribution.
- As a first approximation we can describe the risk of an asset or portfolio by its volatility (standard deviation of returns)

Are Asset Returns Normally Distributed?

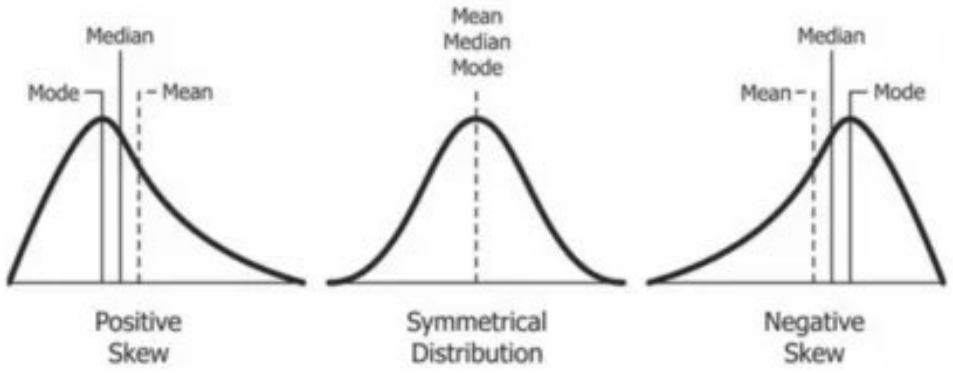
- Thinking of asset returns as being normally distributed is only a model. This model is useful but not the whole truth. Volatility does not capture the full complexity of the distribution



Measuring Higher Moments of Distribution

- Skewness and kurtosis are two statistical measures that provide insights into the shape and distribution of a dataset.

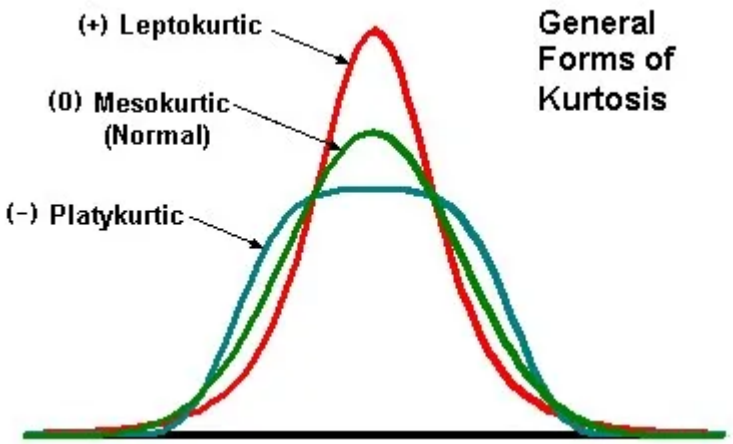
- Skewness measures the asymmetry of a distribution. It indicates whether the data is skewed to the left (negative skew) or to the right (positive skew), or if it has a symmetric distribution (zero skew). Skewness is calculated using the third standardized moment of the data.



Measuring Higher Moments of Distribution

- Skewness and kurtosis are two statistical measures that provide insights into the shape and distribution of a dataset.

- Kurtosis measures the “tailedness” or the degree of outliers present in a distribution. It indicates whether the distribution has heavy tails (leptokurtic) or light tails (platykurtic) compared to a normal distribution.



Tail Risk – Alternative Risk Measure

- Value at Risk (VaR), is perhaps the most used tail risk measure. Here we refer to the nonparametric measure. It is simply the change in wealth associated with a point on the returns probability distribution. Formally it is defined as

$$\mathrm{VaR}_\alpha(x) := -\inf\{x \in \mathbb{R}: F_X(x) > \alpha\}$$

- VaR was formally defined by RiskMetrics in 1994. It is now widely accepted in institutional finance and is required in many reporting regulations. However, a significant shortcoming of VaR is that it is not a 'coherent' measure of risk, defined four criteria that a risk measure must satisfy to be considered 'coherent':
  - Subadditivity
  - Monotonicity
  - Positive Homogeneity
  - Translation Invariance

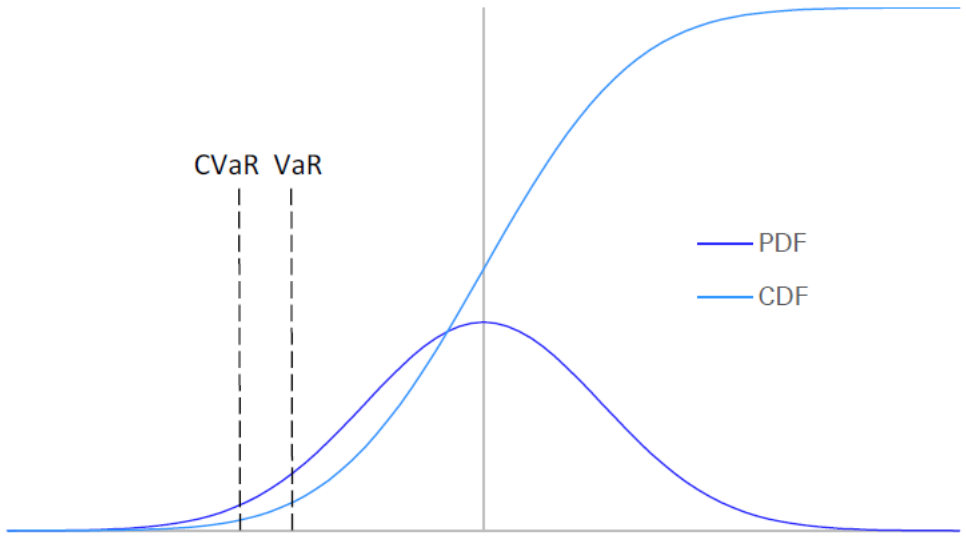
- Of these, perhaps the most important is subadditivity. This condition stipulates that the risk of a portfolio of two assets cannot be greater than the sum of the risks of the two individual assets, which is also known as the principle of diversification.

Tail Risk – Alternative Risk Measure

- A variation on VaR is conditional value at risk (CVaR), which is also known as expected shortfall. This is the expected change in wealth beyond a given loss (VaR) threshold:

$$\mathrm{CVaR}_{\alpha}(x) := -\mathrm{E}[x|x \leq -VaR_{\alpha}(x)]$$

- In contrast to VaR, CVaR is a coherent risk measure. CVaR is gradually gaining more traction in risk management and portfolio construction.



Source: Deutsche Bank Quantitative Strategy

How to Estimate Tail Risk

- Parametric approach
  - Normal distribution:

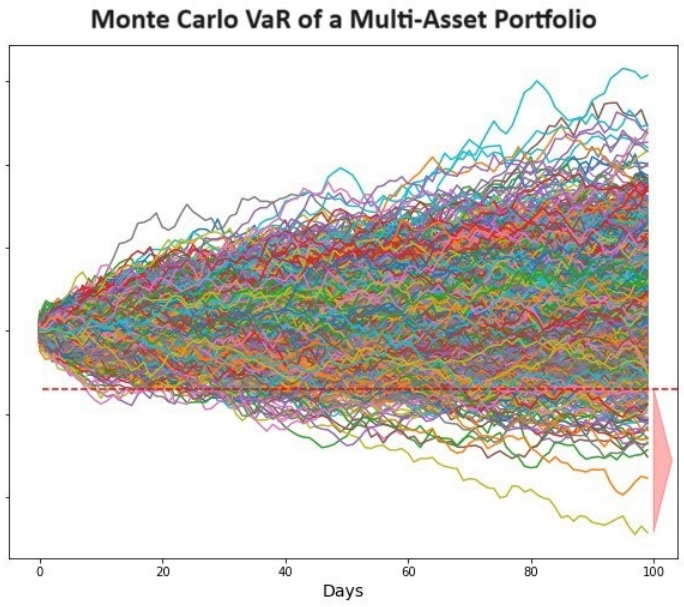
$$VaR_{1-\alpha} = v_\alpha = -z_\alpha = -N^{-1}(\alpha)$$

$$CVaR_{1-\alpha} = -\frac{1}{\alpha}\int_{-\infty}^{z_\alpha}\frac{1}{\sqrt{2\pi}}ze^{-\frac{z^2}{2}}dz = \frac{1}{\alpha}\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{z_\alpha}e^{-\frac{z^2}{2}}d\left(-\frac{z^2}{2}\right) = \frac{1}{\alpha}\frac{1}{\sqrt{2\pi}}\left[e^{-\frac{z^2}{2}}\right]_{-\infty}^{z_\alpha} = \frac{1}{\alpha}\frac{1}{\sqrt{2\pi}}e^{-\frac{z_\alpha^2}{2}} = y_\alpha$$

  - Non-normal distribution: Cornish Fisher Expansion

$$CVaR_{1-\alpha} = Y_\alpha = y_\alpha\left[1 - v_\alpha\frac{S}{6} + (1-2v_\alpha^2)\frac{S^2}{36} + (-1+v_\alpha^2)\frac{K}{24}\right]$$

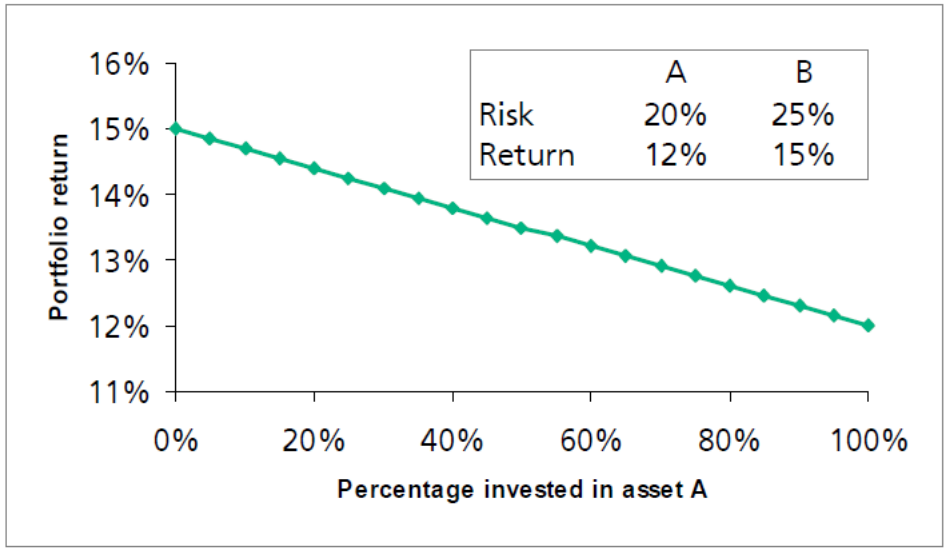
- Non-parametric approach
  - Historical simulation
  - Monte Carlo simulation

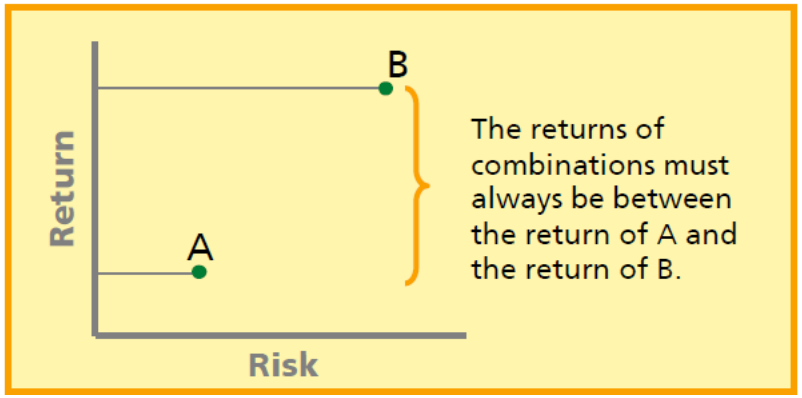


Summary

- All risk measures try to capture risk inherit in true distribution
- Volatility is a robust and well understood measure of risk
  - Volatility measures the width of a distribution
  - Volatility has limitations if the return distribution is not symmetric
  - Volatility has limitations if the return series tend to have many exceptional events
- Alternative risk measures can address the limitations of volatility, but at the cost of complexity

Asset Returns are additive …





… But Risks do not add

- Clearly rA and rB have the same variance, but the sum (rA+rB) is always 8% and hence has zero variance.

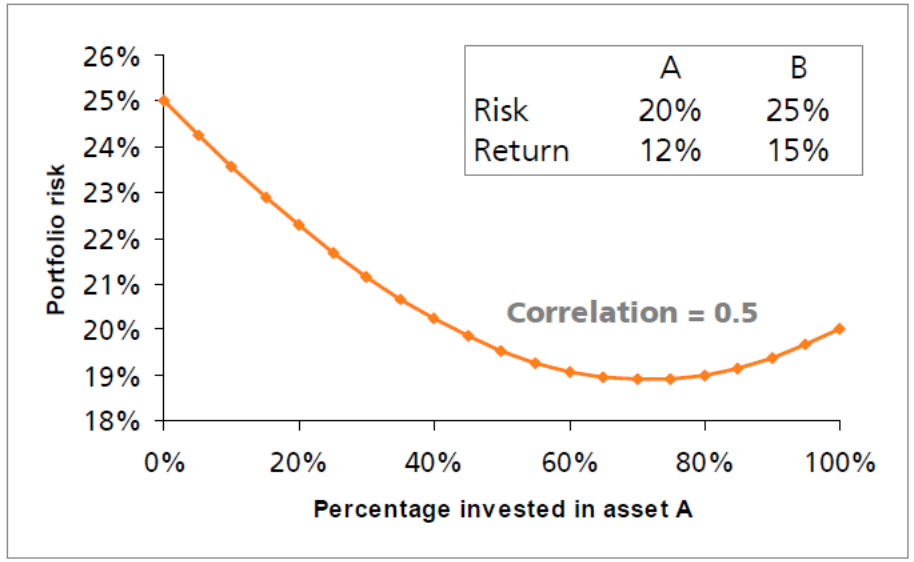
r_A (%)	r_B (%)
1	7
3	5
5	3
7	1

- What is the relationship between the variances of 2 variables and the variance of their sum? We need to specify the co-movements of assets

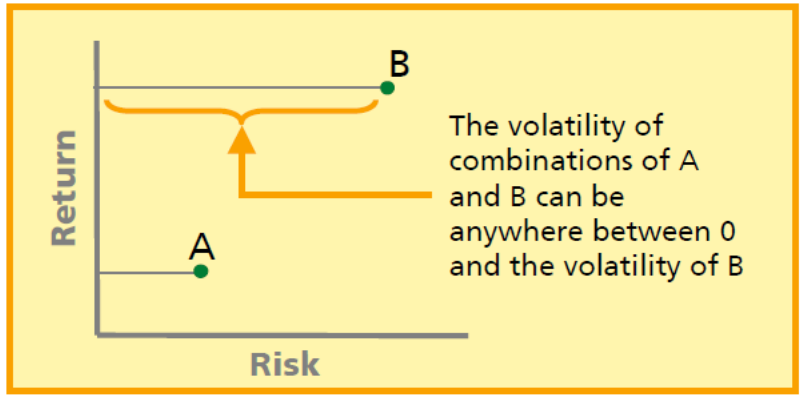
$$\sigma_{x,y} = \frac{1}{N}\sum_t (x_t - \mu_x)(y_t - \mu_y)$$ **Covariance**

$$\rho = \frac{\sigma_{x,y}}{\sigma_x \sigma_y}$$ **Correlation**

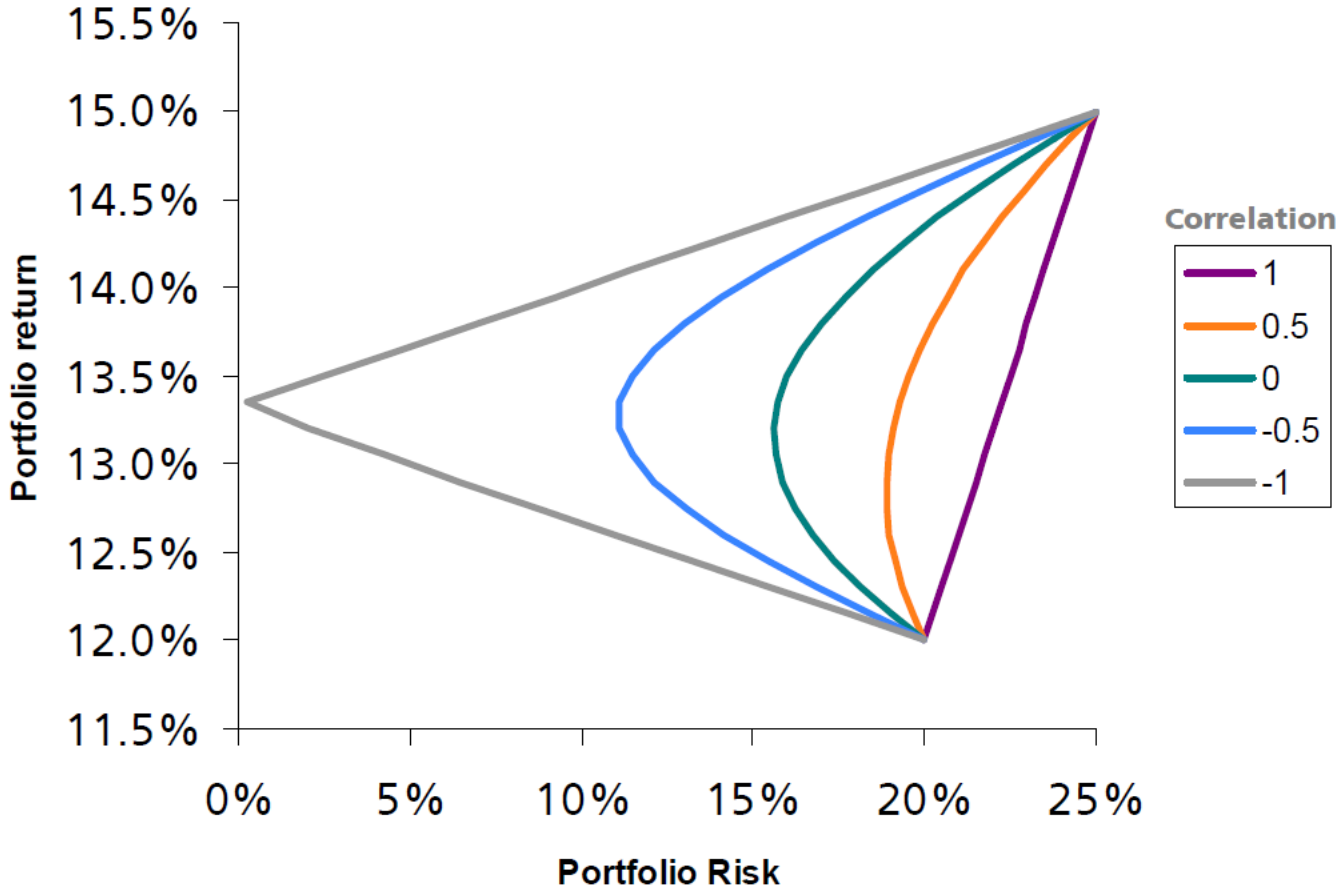
Risk and Correlation



variance(A+B)
=
variance(A)
+
variance(B)
+
2 × covariance(A,B)



The impact of diversification



Measuring diversification

- Asset-based measure: Herfindahl-Hirschman Index (HHI)
  - HHI is a measure of market concentration used to determine concentration of market share in an industry
  - In investment world, it can be used to quantify concentration risk in a portfolio.
  - HHI is calculated by summing the squares of each holding's percentage weights.
  - Lower the value, more diversified is the portfolio
  - HHI measure can extend from holding-based to attribution-based (as long as the inputs sum to 100%)

Measuring diversification

- The range of HHI is from 1/n (equal-weighted portfolio) to 1 (a single asset portfolio). As HHI increases, concentration risk increases.

- The effective number of assets is simply the reciprocal of HHI. It can be used to evaluate whether certain assets may have a disproportionate effect on the performance of the overall portfolio.

Effective number of stocks

Company	Market Share	(Market Share)^2
Bluepoint llc	30.00%	0.0900
Ray Industries	20.00%	0.0400
Powerplay	10.00%	0.0100
Growth Inc	7.50%	0.0056
LTM	7.50%	0.0056
Indigo	5.00%	0.0025
Polaris	4.00%	0.0016
Poseidon Chemical	6.00%	0.0036
Vision Studios	6.00%	0.0036
Citadel	4.00%	0.0016
Hirschman-Herfindahl Index		**0.16**

Effective number of stocks	**6.09**
Actual number of stocks	**10.00**

Measuring diversification

- Risk-based measure: Diversification Ratio
- The diversification ratio DR(w) of a portfolio with asset weights w is the ratio of the weighted average of the asset volatilities to the portfolio volatility, that is

$$DR(w) = \frac{\sigma^t w}{\sqrt{w^t \Sigma w}}$$

- If all assets have correlation 1, there is no diversification in portfolio and DR =1. Hence Diversification Ratio is larger when portfolio diversification is better.

Measuring diversification

- The diversification ratio can be decomposed into an asset correlation component and an asset weight component, and that the diversification ratio increases (resp. decreases) when:
  - ✓The concentration in asset weights decreases (resp. increases)
  - ✓The average correlation between assets decreases (resp. increases)

- The diversification ratio can be interpreted as a correlation-based measure of portfolio diversification consistent with the intuition that portfolios with concentrated weights and/or highly correlated holdings would be poorly diversified, whatever the exact measure of diversification used.

Portfolio Risk and Return

$$\mu_P = \sum_i w_i \mu_i$$

$$\sigma_P^2 = \sum_i w_i^2 \sigma_i^2 + \sum_{i \neq j} \sigma_{ij} w_i w_j$$

$$= \sum_{i,j} w_i \sigma_{ij} w_j \quad (\text{where } \sigma_{ii} = \sigma_i^2)$$

$$\mu_P = \mathbf{w}' \cdot \boldsymbol{\mu}, \qquad \sigma_P^2 = \mathbf{w}' \cdot \mathbf{V} \cdot \mathbf{w}$$

The sign ' denotes matrix transposition

$$\mathbf{V} = \begin{pmatrix} \sigma_1^2 & \sigma_{1,2} & \cdots & \sigma_{1,N} \\ \sigma_{2,1} & \sigma_2^2 & & \vdots \\ \vdots & & \ddots & \sigma_{N-1,N} \\ \sigma_{N,1} & \cdots & \sigma_{N,N-1} & \sigma_N^2 \end{pmatrix} \qquad \mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_N \end{pmatrix} \qquad \boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_N \end{pmatrix}$$

$$\mathbf{w}' = \begin{pmatrix} w_1 & w_2 & \cdots & w_N \end{pmatrix}$$

Portfolio Risk Attribution

- Which holdings are the main contributors to portfolio risk?
- Risk is not linear – can not decompose risk by holdings
- But we can do a marginal risk analysis
- How much does a small change in the portfolio weights affect the portfolio risk?
- Marginal contribution to total risk is the (marginal) change in Portfolio risk due to a (marginal) change in portfolio weights

Marginal contribution to risk

- $MCTR_i$ denotes the marginal contribution to total risk of the i-th asset

- For an **unconstrained** portfolio we can show that

$$\boxed{\mathbf{MCTR} = \frac{\partial \sigma_P}{\partial \mathbf{w}} = \frac{1}{\sigma_P} \cdot \mathbf{V} \cdot \mathbf{w}} \qquad \mathbf{MCTR} = \begin{pmatrix} MCTR_1 \\ \vdots \\ MCTR_N \end{pmatrix}$$

- This can be written as

$$\boxed{\mathbf{MCTR} = \boldsymbol{\beta} \cdot \sigma_P} \qquad \boldsymbol{\beta} = \frac{1}{\sigma_P^2} \cdot \mathbf{V} \cdot \mathbf{w} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_N \end{pmatrix}$$

where β_i is the beta of the i-th asset relative to the portfolio

Marginal contribution to risk - Example

- Consider again the previous portfolio (correlation = 0.5)
- With 40% in Asset A and 60% in asset B, the portfolio risk is 20.2%
- If we increase the weight of asset A to 41% then the portfolio risk increases by 15.3 basis points
- If we increase the weight of asset B to 61% then the portfolio risk increases by 23.5 basis points
- If we increase the weight of asset A to 41% and decrease the weight of asset B to 59% then the portfolio risk will decrease by 8.2 basis points (8.2 = 23.5 - 15.3)

$$\mathbf{w} = \begin{pmatrix} 0.4 \\ 0.6 \end{pmatrix} \qquad \sigma_P = 20.22\% \qquad \mathbf{V} = \begin{pmatrix} 4\% & 2.5\% \\ 2.5\% & 6.25\% \end{pmatrix}$$

$$\mathbf{V} \cdot \mathbf{w} = \begin{pmatrix} 3.10\% \\ 4.75\% \end{pmatrix} \qquad \mathbf{MCTR} = \begin{pmatrix} 15.3\% \\ 23.5\% \end{pmatrix} \qquad \boldsymbol{\beta} = \begin{pmatrix} 0.758 \\ 1.161 \end{pmatrix}$$

Marginal contribution to risk

- By how much does the portfolio risk increase if we increase the weight in asset A to 40.4% and the weight in asset B to 60.6%?
- The weighted MCTR is the MCTR multiplied by the relative asset weights
- The weighted MCTR for the i-th asset is therefore wi x MCTRi
- The weighted MCTR is sometimes regarded as a decomposition of the portfolio risk

$$\boxed{w_1 \cdot MCTR_1 + w_2 \cdot MCTR_2 + \cdots + w_N \cdot MCTR_N = \sigma_P}$$

Marginal contribution to tail risk

- A traditional parametric approach is based on statistical distribution assumption (such as normal distribution), which usually has limited applications. A non-parametric method is needed for estimating quantile-based risk attribution (marginal CVaR).

- The portfolio CVaR (expected loss), which is defined as

$$CVaR_C = E\left[R | R \leq VaR_c = Q_R(1-c)\right],$$

where $R = \vec{\alpha}'\vec{R} = \sum_{i=1}^{N} \alpha_i R_i$, $\vec{R}$ is forecasting returns of the assets, $\vec{\alpha}$ represents the respective portfolio weights, $Q_R(1-c)$ denotes the (1-c)*100–quantile of the distribution of R

Marginal contribution to tail risk

- Because CVaR is homogeneous of degree one, it satisfies the relation

$$CVaR_c = \sum_{i=1}^{N} \alpha_i \frac{\partial CVaR_c}{\partial \alpha_i} \ ,$$

which represents a marginal decomposition. Thus, the marginal contribution of *ith* component to the portfolio CVaR is

$$MCVaR_i = \alpha_i \frac{\partial CVaR_c}{\partial \alpha_i}.$$

Marginal contribution to tail risk

- The "per-unit" marginal contribution $\frac{\partial CVaR_c}{\partial \alpha_i}$ can be expressed conveniently as in Scaillet (2002):

$$\frac{\partial CVaR_c}{\partial \alpha_i} = \mathrm{E}[R_i | R \leq VaR_c].$$

which implies the marginal CVaR is the expected loss per unit of component *i,* given that the portfolio loss less than or equals the VaR under some scenarios.

Marginal contribution to tail risk

- If all the components of a portfolio are elliptically distributed, CVaR and its marginal contribution can be calculated analytically, given estimated expected return and covariances, as shown by Meucci (2005). However, the assumption of elliptically distributed market is very restrictive. Under the non-elliptically distribution framework, no analytical estimation for CVaR or marginal CVaR exists.

- The scenario extraction method, independent of return distribution assumption, can be applied to marginal CVaR estimation. It is known that the empirical estimator for CVaR, when the number of simulations S is large, is a weighted average of multiple order statistics

$$CVaR_c = \sum_{k=1}^{S} \omega_{S,k} R_{(k)}, \ where \ \ \omega_{S,k} = \begin{cases} \frac{1}{K}, k \leq K \\ 0, k > K \end{cases}, \qquad \frac{K-1}{S} < 1 - c \leq \frac{K}{S}, \qquad (5)$$

where $R_{(k)}$ is the kth order statistics of the simulated portfolio returns.

Marginal contribution to tail risk

- The "per-unit" marginal CVaR contribution $\frac{\partial CVaR_c}{\partial \alpha_i}$ can be simplified as (see Mausser 2001, Meucci 2005):

$$\frac{\partial CVaR_c}{\partial \alpha_i} = \sum_{k=1}^{S} \omega_{S,k} R_{i,(k)} \ , \qquad (6)$$

where $\omega_{S,k}$ is a step function that jumps from 0 to $\frac{1}{K}$ as described in Equation (5) and $R_{i,(k)}$ denotes the return of component *i* in the scenario that results in the *k*th largest portfolio loss from simulations.

Marginal contribution to tail risk

- The scenario extraction method gives more flexibility than the parametric method, which depends on simulated scenarios instead of distribution assumption. It also gives more accurate and less noisy estimation since it is based on a weighted average of a range of order statistics that has an inherent smoothing effect.
- In terms of computation, scenario extraction approach is highly efficient for portfolio risk decomposition, particularly where the portfolio contains large number of assets.

- Reference

Mausser, H., 2001, Calculating quantile-based risk analytics with L-estimators, ALGO Research Quarterly, Vol.4 No. 4, 33-48.

Scaillet, O., 2002, Nonparametric estimation and sensitivity analysis of expected shortfall. *Mathematical Finance* **14**(1), 74–86.

Meucci, A., 2005, Risk and Asset Allocation (Springer).

Broad Types of Portfolio Risk

- **Market Risk**: Aptly named, this risk stems from the market's inherent volatility, casting a shadow over portfolios. Within this realm of market risk, distinct manifestations take form: equity risk, interest rate risk, and currency risk. Three sub-categories define this risk:
  - **Equity Risk**: The risk stemming from fluctuations in stock market prices. It materializes as the unsettling prospect of losing capital in the wake of a stock market decline. Investments intertwined with the stock market ride the tide of its fluctuations, potentially leading to financial erosion.
  - **Interest Rate Risk**: The risk related to changes in interest rates, particularly relevant for debt investments. Debt investments are particularly susceptible, as alterations in interest rates can trigger losses or gains.
  - **Currency Risk**: This relates to the risk of loss due to variations in foreign exchange rates. A shifting exchange rate can diminish returns when converted back to the investor's currency.

Broad Types of Portfolio Risk

- **Liquidity Risk**: This risk arises when an investor is unable to sell an investment quickly without a substantial loss in value. Liquidity Risk can be detrimental in emergency situations where cash is needed promptly. Investments are held hostage by low market values.

- **Concentration Risk**: Concentration Risk is the risk associated with investing heavily in a single asset type. The peril here lies in channeling all financial resources into a solitary asset category, like the stock market. Diversification across different asset classes can help in mitigating this risk and shows the importance of a well-balanced investment portfolio.

- **Credit Risk**: In the context of bonds, credit risk is concerned with the financial stability of the bond issuer. It manifests when the issuer of bonds grapples with financial troubles, jeopardizing the return of investment. Caution is a virtue; checking credit ratings of bonds before investment can help in understanding this risk better.